

# CREATIVE INSTITUTE DATA SCIENCE

## Data Cleaning

### Practical -3 Measure of Central Tendency for each class

Finding null values in row for particular columns.

we have find 3 variables .

for find class, need to knowledge about domain expertise..

in this data set we clearly shows that we have 3 missing values variable

**LotFrontage, MassvnrArea, and GarrageYrBl**

int this variable , see in dataset LotFrontage is related to

LotArea which has value not category

LotShape which has category values and

LotConfig which is also category data

We take LotConfig for reffrence class to fill values

Now we need to find out how many classes in lotconfiging has using **unique**

```
df['LotConfig'].unique()
```

```
['Inside', 'FR2', 'Corner', 'CulDSac', 'FR3']
```

There are commond categorical unique values.

Now we have to find out the rows which has "Inside" data

```
df[df.loc[:, 'LotConfig']=="Inside"]
```

there is all the columns present which rows present in "Inside"

now we need only LotFrontage columns which rows has Inside for that

```
df[df.loc[:, 'LotConfig']=="Inside"]['LotFrontage']
```

or we can store in perticular variable

```
df4=df[df.loc[:, 'LotConfig']=="Inside"]['LotFrontage']  
df4
```

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for that we have all the values which has "Inside" in rows of LotFrontage columns in this LotFrontage values we have some None values available . we fill missing values to replace none values to mean values

```
df4.replace(np.nan,df4.mean())
```

this is only for one category data we fill missing values but we have bunch of data for more category

we need to fill each categorical data to fill missing values in each category.

We need to create copy of original data set and update this copied dataset.

```
df_copy=df.copy()
for var_class in df['LotConfig'].unique():

    df_copy.update(df[df.loc[:, 'LotConfig']==var_class]['LotFrontage']
                    .replace(np.nan,df[df.loc[:, 'LotConfig']==var_class]['LotFrontage'].mean()))
```

Now find out there is any missing value present in LotFrontage class

```
df_copy.isnull().sum()
```

we clearly see there is not null values present in LotFrontage class.

Now we need to fill values in another columns also

For all the missing columns available we need to find domain expertise to

Related categorical columns

We have three columns= ["LotFrontage","MassVnrArea","GarageYrBlt"]

For this three columns we select related columns like

["LotConfig","MassVnrType","GarageType"]

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```
num_vars=['LotFrontage', 'MasVnrArea', 'GarageYrBlt']  
cat_vars=['LotConfig', 'MasVnrType', 'GarageType']
```

for select multiple columns

```
df_copy=df.copy()  
num_vars=['LotFrontage', 'MasVnrArea', 'GarageYrBlt']  
cat_vars=['LotConfig', 'MasVnrType', 'GarageType']  
  
for cat_var, num_var in zip(cat_vars,num_vars):  
  
    for var_class in df[cat_var].unique():  
  
        df_copy.update(df[df.loc[:,cat_var]==var_class][num_vars]  
                        .replace(np.nan,df[df.loc[:,cat_var]==var_class][num_vars].mean()))
```

Now check there is any null value available or not

```
df_copy[num_vars].isnull().sum()
```

so there isn't any null value available..

now see data distributioin

```
plt.figure(figsize=(10,10))  
sns.set()  
for i ,j in enumerate(num_vars):  
    plt.subplot(2,2,i+1)  
    sns.distplot(df[j],bins=20,kde_kws={'linewidth':8,'color':'red'},label='original')  
    sns.distplot(df_copy[j],bins=20,kde_kws={'linewidth':5,'color':'green'},label='mean')  
    plt.legend()
```

